Why quantum mechanics?

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Fondation Danoise

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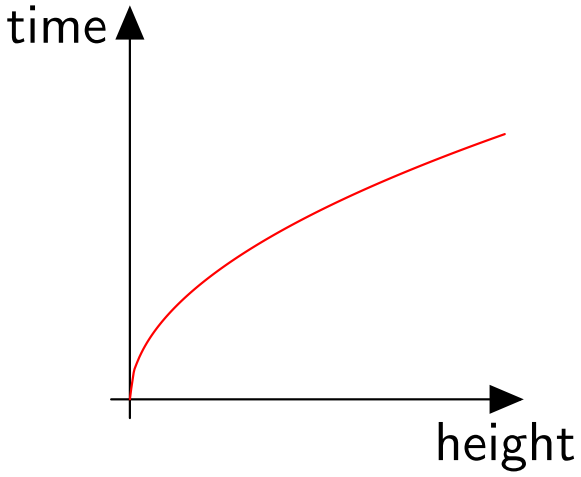

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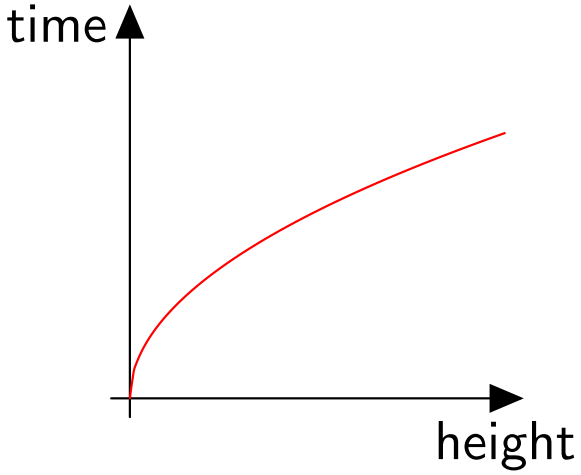
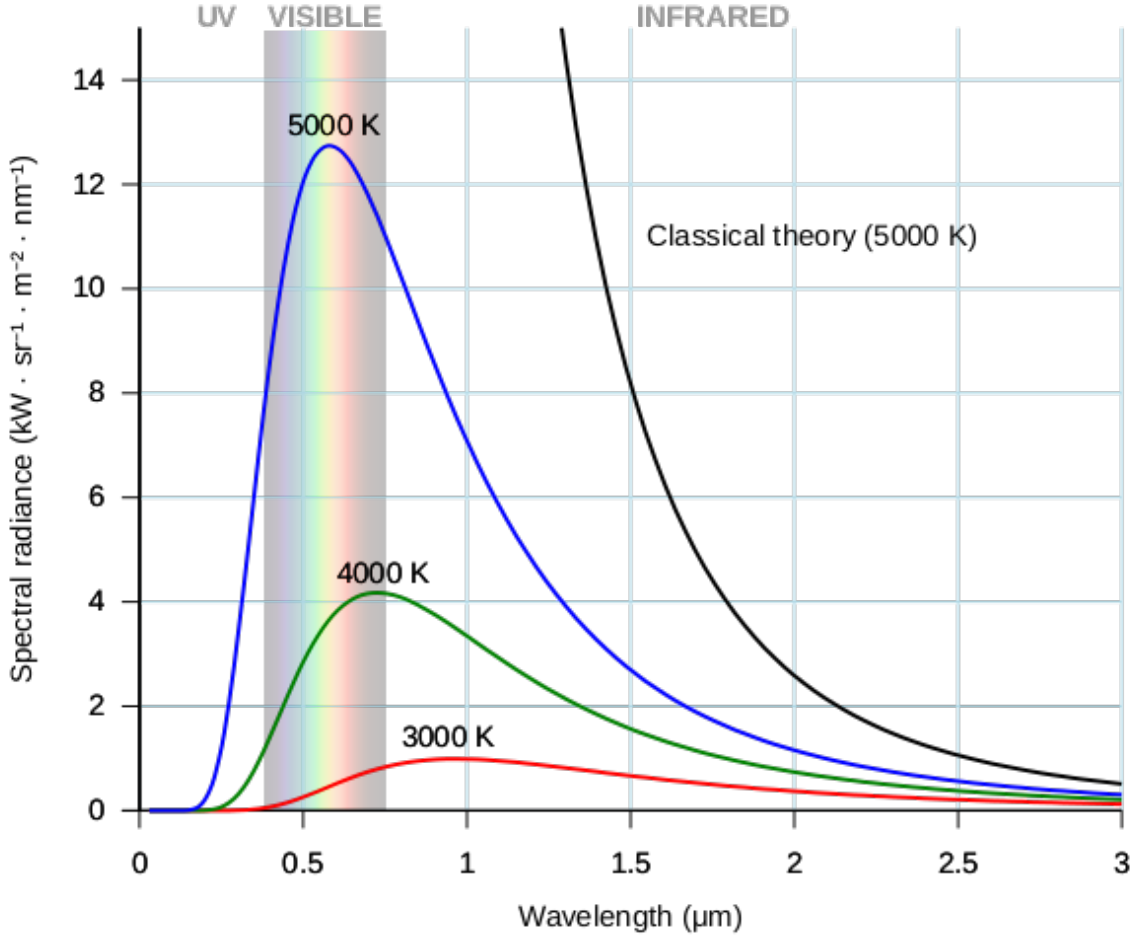


- Explanation: Newton's law of gravity

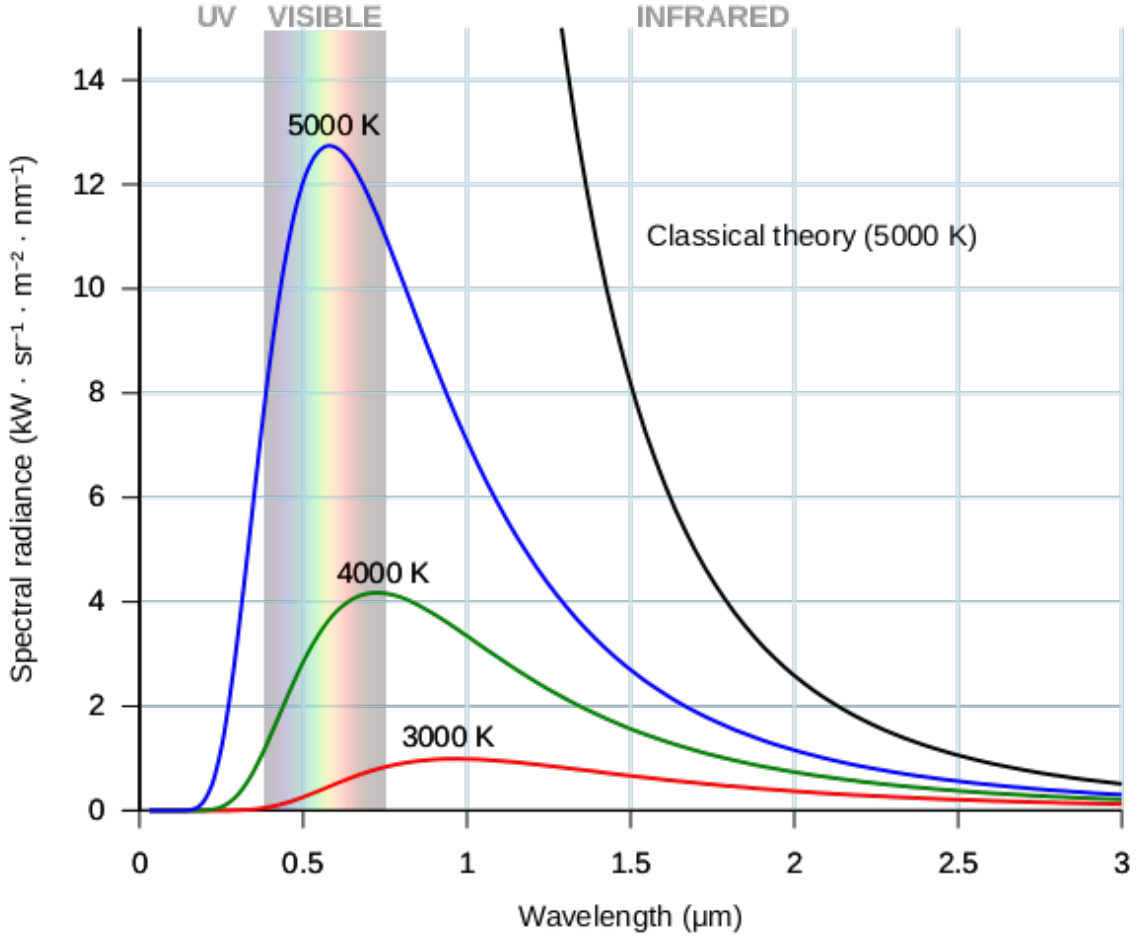
Black-body radiation



Wavelength dependence of radiation from black-body (e.g. the Sun, an incadecent light bulb)

Ultraviolet catastrophe

Black-body radiation



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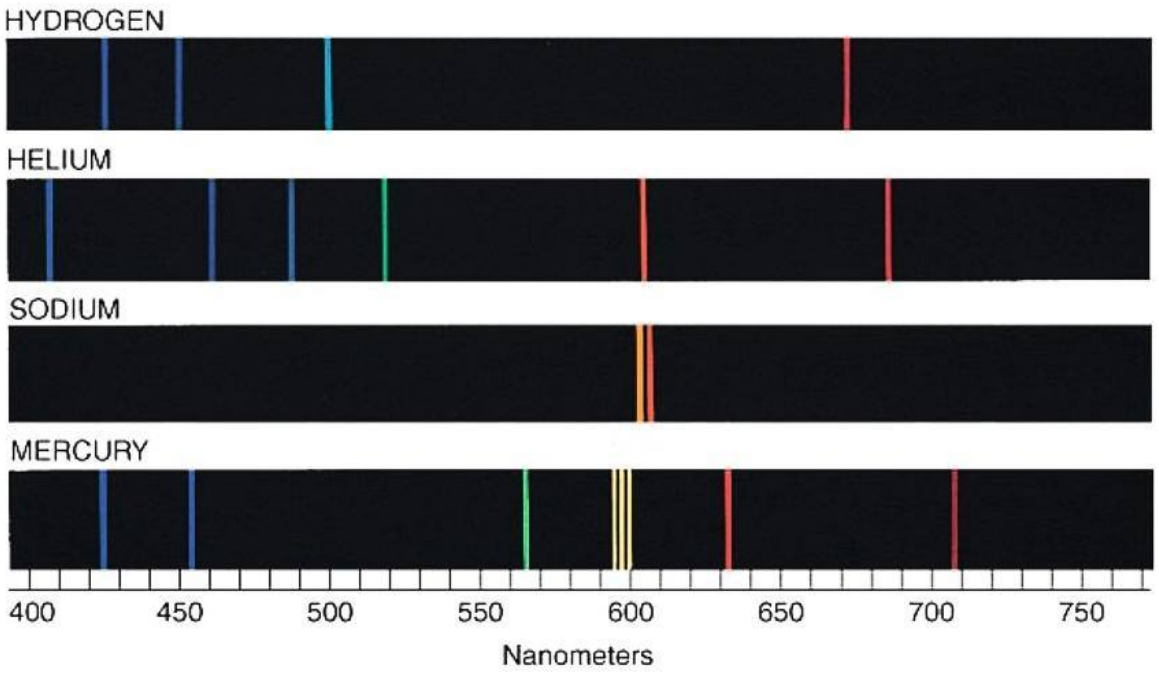
Solved by Planck radiation law (1900). **Quantized energies.**

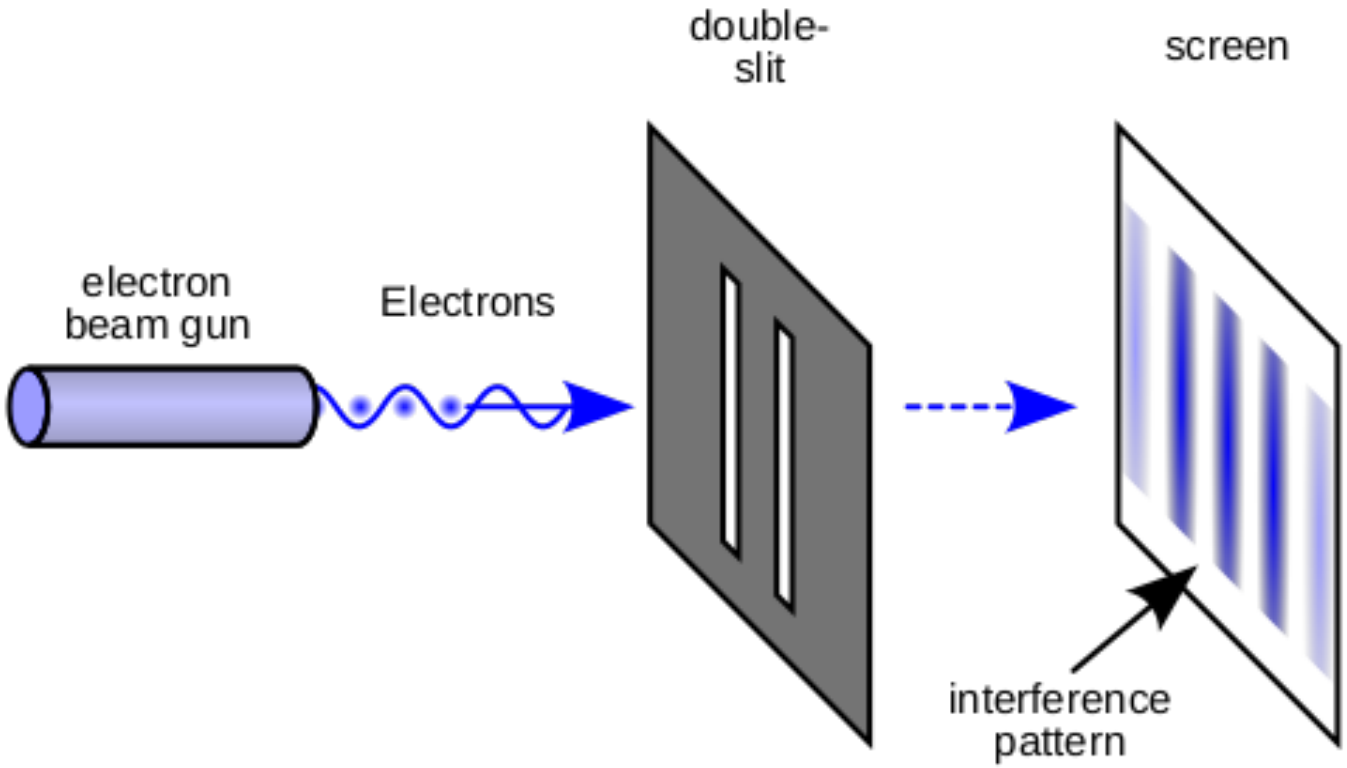
Atomic emission spectra



Spectrum of hydrogen explained by the Bohr model of the atom (1913). **Quantized energies.**

Spectrum of larger atoms from theory of random matrices, pioneered by Wigner (1955)

Double slit experiment



Light: Young (1801)
Light behaves like a wave

Electrons: Davidson–Germer and Thomson–Reid (1927)
Electrons behave like a wave

Quantum today

- 2025 is UNESCO "international year of quantum science and technology"

INTERNATIONAL YEAR OF
Quantum Science
and Technology

- Quantum computing
- Clocks (important for e.g. GPS)
- Molecular chemistry — properties of molecules
- LED's
- ...